(iii) Stars with $0.08\ \mathbf{M}_{\odot} < m < 0.26\ \mathbf{M}_{\odot}$ are completely convective, leading to hydrogen burning to helium, until all hydrogen is exhausted, after which the star contracts to a *white dwarf.*

Other stars, with $m > 0.26\ \mathbf{M}_{\odot}$, evolve differently depending on their masses, as shown in Figure 12.4 and described in the following sections.

12.4. FUSION PROCESSES IN STARS

The elemental composition of our sun is about 73% hydrogen, 25% helium, and 2% carbon, nitrogen, oxygen, and other elements distributed as shown in Figure 12.5. In all, approximately 70 elements have been detected in the solar spectrum and there are reasons to believe that all the elements to uranium are present in our sun. Let us now consider the reactions for the formation of all these elements and the energy producing nuclear processes in our sun and other stars.

The present understanding of processes in the interior of stars is the result of combined efforts from many scientific disciplines such as hydrodynamics, plasma physics, nuclear physics, nuclear chemistry and not least astrophysics. To understand what is going on in the inaccessible interior of a star we must make a model of the star which explains the known data: mass, diameter, luminosity, surface temperature and surface composition. The development of such a model normally starts with an assumption of how elemental composition varies with distance from the center. By solving the differential equations for pressure, mass, temperature, luminosity and nuclear reactions from the surface (where these parameters are known) to the star's center and adjusting the elemental composition model until zero mass and zero luminosity is obtained at the center one arrives at a model for the star's interior. The model developed then allows us to extrapolate the star's evolution backwards and forwards in time with some confidence. Figure 12.7 shows results from such modelling of the sun.

12.4.1. Hydrogen burning to helium

Helium can be formed from hydrogen in several ways, the least likely one is (12.5), which would require that 4 protons come together simultaneously. Similarily to normal chemical rate reactions a reaction of this type involving a multi body reaction it is likely that the reaction is a multi-step process which one rate determining step. The possible multi-step processes are (not showing γ's)

$$^{1}\mathrm{H} + {}^{1}\mathrm{H} \rightarrow {}^{2}\mathrm{H} + \left|\mathrm{e}^{-} + \mathrm{e}^{+}\right| + \nu_{\mathrm{e}} \quad Q = 0.42 + 1.02 = 1.44\ \mathrm{MeV} \tag{12.13}$$

$$^{1}\mathrm{H} + {}^{2}\mathrm{H} \rightarrow {}^{3}\mathrm{He} \qquad Q = 5.49\ \mathrm{MeV} \tag{12.15}$$

$$^{1}\mathrm{H} + {}^{3}\mathrm{H} \rightarrow {}^{4}\mathrm{He} \qquad Q = 19.81\ \mathrm{MeV} \tag{12.16}$$

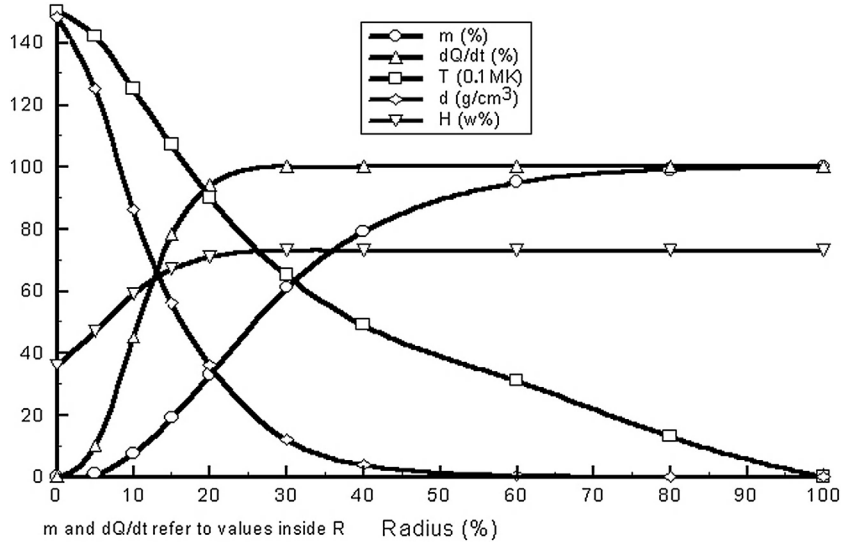


Figure 12.7 Calculated composition of our sun (mass, energy production, temperature, density and hydrogen content) as function of the radius.

$$^{2}\mathrm{H} + \mathrm{n} \rightarrow {}^{3}\mathrm{H} \qquad Q = 6.25\ \mathrm{MeV} \qquad (12.17)$$

$$^{2}\mathrm{H} + {}^{2}\mathrm{H} \rightarrow {}^{3}\mathrm{H}\ (1.01\ \mathrm{MeV}) + {}^{1}\mathrm{H}\ (3.02\ \mathrm{MeV}) \qquad Q = 4.03\ \mathrm{MeV} \qquad (12.18)$$

$$^{2}\mathrm{H} + {}^{2}\mathrm{H} \rightarrow {}^{3}\mathrm{He}\ (0.82\ \mathrm{MeV}) + \mathrm{n}\ (2.45\ \mathrm{MeV}) \qquad Q = 3.27\ \mathrm{MeV} \qquad (12.19)$$

$$^{2}\mathrm{H} + {}^{2}\mathrm{H} \rightarrow {}^{4}\mathrm{He} \qquad Q = 23.85\ \mathrm{MeV} \qquad (12.14)$$

$$^{2}\mathrm{H} + {}^{3}\mathrm{H} \rightarrow {}^{4}\mathrm{He}\ (3.52\ \mathrm{MeV}) + \mathrm{n}\ (14.06\ \mathrm{MeV}) \qquad Q = 17.58\ \mathrm{MeV} \qquad (12.20)$$

$$^{2}\mathrm{H} + {}^{3}\mathrm{He} \rightarrow {}^{4}\mathrm{He}\ (3.67\ \mathrm{MeV}) + {}^{1}\mathrm{H}\ (14.67\ \mathrm{MeV}) \qquad Q = 18.35\ \mathrm{MeV} \qquad (12.21)$$

$$^{3}\mathrm{H} + {}^{3}\mathrm{H} \rightarrow {}^{4}\mathrm{He}\ (3.78\ \mathrm{MeV}) + 2\mathrm{n}\ (7.55\ \mathrm{MeV}) \qquad Q = 11.33\ \mathrm{MeV} \qquad (12.22)$$

$$^{3}\mathrm{He} + 2\mathrm{n} \rightarrow {}^{4}\mathrm{He}\ (4.27\ \mathrm{MeV}) + \mathrm{n}\ (17.09\ \mathrm{MeV}) \qquad Q = 21.36\ \mathrm{MeV} \qquad (12.23)$$

$$^{3}\mathrm{He} + {}^{3}\mathrm{He} \rightarrow {}^{4}\mathrm{He}\ (4.29\ \mathrm{MeV}) + 2\,{}^{1}\mathrm{H}(8.57\ \mathrm{MeV}) \qquad Q = 12.86\ \mathrm{MeV} \qquad (12.24)$$

All reactions are exothermic. Figure 12.8 shows how reaction cross sections increase with temperature. As the number of particles of a particular energy decreases with energy

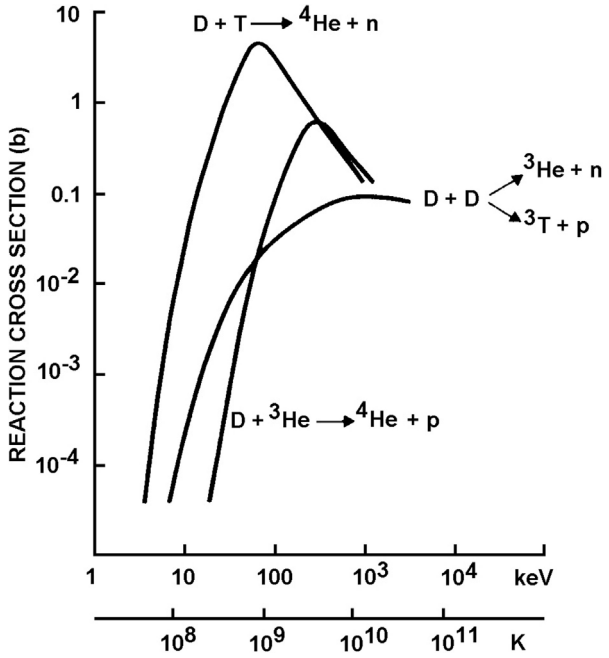


Figure 12.8 Reaction cross sections for some light element fusion reactions at solar conditions. *(From Bowler.)*

(Fig. 12.6) this results in a maximum number of reactions at a certain energy, E_{max}. The fastest hydrogen fusion reaction is (12.20), which, however, requires that tritium has been synthesized in an earlier step, e.g. by (12.17) or (12.18).

(i) For stars with $m \leq 1.5\ \mathbf{M}_{\odot}$ and $T \geq 2 \times 10^7$ K, the main reaction sequence is (12.13) followed by (12.15) and (12.24). This is referred to as the *proton-proton chain*, summarized in Table 12.2. Reaction times in Tables 12.2 and 12.3 are from Gamow and refer to solar conditions.

In 9% of the pp-chain, reaction (12.21) is replaced by

$$^3\mathrm{He} + {}^4\mathrm{He} \rightarrow {}^7\mathrm{Be} \qquad Q = 1.59\ \mathrm{MeV} \tag{12.25}$$

$$^7\mathrm{Be} + \mathrm{e}^- \rightarrow {}^7\mathrm{Li} + \nu_e(4\%) \qquad Q = 1.37\ \mathrm{MeV} \tag{12.26}$$

Table 12.2 The proton-proton chain for ^{4}He formation; about 90% of the solar energy production

Reaction steps	*Q* (MeV)	Mean reaction time	Eqn.
$^1\mathrm{H} + {}^1\mathrm{H} \rightarrow {}^2\mathrm{H} + \mid \mathrm{e}^- + \beta^+ \mid + \nu$ (2%)	1.44 MeV	1.4×10^{10} y	(12.13)
$^2\mathrm{H} + {}^1\mathrm{H} \rightarrow {}^3\mathrm{He} + \gamma$	5.49 MeV	6 s	(12.15)
$^3\mathrm{He} + {}^3\mathrm{He} \rightarrow {}^4\mathrm{He} + 2\ {}^1\mathrm{H}$	12.86 MeV	9×10^5 y	(12.24)
Overall $4\ {}^1\mathrm{H} \rightarrow {}^4\mathrm{He} + \mid 2\mathrm{e}^- + 2\beta^+ \mid + 2\nu$	26.7 MeV	1.4×10^{10} y	(12.6)

Table 12.3 The CNO cycle for helium production; about 10% of the solar energy

Reaction steps	*Q* (MeV)	Mean reaction time	eqn.
$^{12}C + {}^{1}H \rightarrow {}^{13}N + \gamma$	1.94	10^7 y	(12.41)
$\downarrow$			
$^{13}C + \lvert e^- + \beta^+ \rvert + \nu$	1.20	7 min	(12.42)
$^{13}C + {}^{1}H \rightarrow {}^{14}N + \gamma$	7.55	3×10^6 y	(12.43)
$^{14}N + {}^{1}H \rightarrow {}^{15}O + \gamma$	7.29	3×10^8 y	(12.44)
$\downarrow$			
$^{15}N + \lvert e^- + \beta^+ \rvert + \nu$	1.74	2 min	(12.45)
$^{15}N + {}^{1}H \rightarrow {}^{12}C + {}^{4}He$	4.86	10^5y	(12.46)
Overall $4\ {}^{1}H \rightarrow {}^{4}He + \lvert 2e^- + 2\beta^+ \rvert + 2\ \nu$	26.7	3×10^8 y	(12.6)

$$^{7}Li + {}^{1}H \rightarrow 2\,{}^{4}He \qquad Q = 17.34\ \text{MeV} \qquad (12.27)$$

The neutrino takes 4% of the decay energy. To a very small amount (<1%) also the reaction sequence

$$^{7}Be + {}^{1}H \rightarrow {}^{8}B; \quad {}^{8}B + e^- \rightarrow {}^{8}Be \rightarrow 2\,{}^{4}He \quad Q = 18.21\ \text{MeV} \qquad (12.28)$$

occurs. Thus isotopes of Li, B and Be are formed as intermediates. ^{7}Be decays by electron capture (see Fig. 5.7) while ^{7}Li is stable. ^{8}B is very shortlived, decaying to ^{8}Be which immediately decays into 2 He. The number of p-p-fusion reactions in our sun amounts to $1.8 \times 10^{38}\ s^{-1}$. The solar neutrino emission comes to ~15% from reaction (12.26) (E_ν 0.38 − 0.86 MeV) and to ~85% from (12.13) ($E_\nu \leq 0.4$ MeV).

(ii) For stars with $m \geq 1.4\ \mathbf{M}_\odot$ Bethe and Weizsäcker in the 1930s deduced the so-called "*CNO*-" or *carbon cycle*, Table 12.3. In such stars temperature and pressure reach higher values, and the consumption of hydrogen is faster. A star of 20 $\mathbf{M}_\odot$ burns its hydrogen through the CNO-cycle in some 10 My, compared to the suns pp-cycle, which burns hydrogen at a slower rate for about 10 000 My.

The CNO-cycle requires the presence of some ^{12}C, which acts as a catalyst. In a hydrogen burning star some small amounts of ^{12}C is always produced through reaction (12.29).

In our sun ~6% of the hydrogen originally in the core has now been burnt. Since helium has a greater mass than hydrogen, it accumulates in the core, while most of the hydrogen burning moves to a layer around the He. As the hydrogen fuel is consumed, the temperature of the core decreases somewhat. However, as the amount of helium is increased and the hydrogen depleted, the core contracts through the increased

gravitational attraction of the helium, and the temperature of the core rises. This heats the outer layers of hydrogen and results in an expansion of the outer mantle of the star, which in turn results in a cooler surface, so that the star irradiates more red light. The star at this period of its life is referred to as a *Red Giant*, see Figure 12.4. We can expect that our sun in about five eons from now will pass through a red giant stage at which time its diameter should expand sufficiently to engulf the inner planets of the solar system.

12.4.2. Helium-burning to iron

(i) For stars with $0.26\ \mathbf{M}_{\odot} < m < 1.5\ \mathbf{M}_{\odot}$, the contracting helium core degenerates and warms. When a density of $\sim 10^7\ \mathrm{kg\,m^{-3}}$, and a temperature of $\sim 1.5 \times 10^8\ \mathrm{K}$ are reached, He begins to fuse. Even though $^{8}\mathrm{Be}$ has an extremely short lifetime, there is always a small equilibrium amount present. It has been calculated that in some red giants one $^{8}\mathrm{Be}$ nucleus is in equilibrium with $\sim 10^9$ nuclei of $^{4}\mathrm{He}$. This amount is sufficient to allow some capture of a third helium nucleus to form $^{12}\mathrm{C}$. The reaction is

$$^{8}\mathrm{Be} + {}^{4}\mathrm{He} \rightarrow {}^{12}\mathrm{C} \tag{12.29}$$

also referred to as the "3α-process" as 3 He form a C atom. This produces a sudden heating at the center, and in fact the core can be considered to detonate, the *helium flash* (Fig. 12.4), though this detonation is dampened by the large outer masses. In such a star helium burning goes on in the core and hydrogen burning at the outer shell. As the central helium is exhausted, helium burning proceeds in an outer shell, while carbon collects in the center. For our sun the He-burning period is expected to last for only 60 My after which ~50% of the core will be C and O. The fusion reactions will then halt. The outer envelope, which through turbulent mixing includes some core material, expands and the star loses some of its mass. The expanding envelope forms a *planetary nebula* with the star in the center termed a white dwarf, see Figure 12.4.

(ii) Stars of masses $>1.5\ \mathbf{M}_{\odot}$ burn hydrogen through the CNO cycle. This requires that the core be convective. The main sequence phase ends as the hydrogen in the core is exhausted, and shell burning begins. The helium core remains convective and non-degenerate, and helium burning begins without perturbations, leading to the formation of mainly carbon and oxygen.

(iia) Stars with $3\ \mathbf{M}_{\odot} < m < 15\ \mathbf{M}_{\odot}$ burn hydrogen and helium in outer shells. For stars of masses $>3.5\ \mathbf{M}_{\odot}$ helium burning becomes the important energy source. Once $^{12}\mathrm{C}$ has been formed, further reactions with helium can explain the formation of oxygen, neon and higher elements according to

$$^{12}\mathrm{C}\,(^{4}\mathrm{He},\gamma)\,^{16}\mathrm{O}(^{4}\mathrm{He},\gamma)\,^{20}\mathrm{Ne}\,(^{4}\mathrm{He},\gamma)\,^{24}\mathrm{Mg}\,(^{4}\mathrm{He},\gamma)\,^{28}\mathrm{Si} \tag{12.30}$$

These reactions occur with increasing yields in stars of increasing mass.

(iib) Carbon fusion can occur in stars $>7.5\ \mathbf{M}_\odot$ and at core temperatures $\geq 8 \times 10^8$ K:

$$\begin{aligned} {}^{12}\mathrm{C} + {}^{12}\mathrm{C} &\rightarrow {}^{24}\mathrm{Mg} \\ &\rightarrow {}^{23}\mathrm{Na} + {}^{1}\mathrm{H} \\ &\rightarrow {}^{20}\mathrm{Ne} + {}^{4}\mathrm{He} \end{aligned} \tag{12.31}$$

This occurs suddenly, and is observed as a *carbon flash*, Figure 12.4. The star then either continues to burn carbon, or explodes (a *supernova*) with destruction of most of the star.

(iic) In the very heavy stars, $m > 15\ \mathbf{M}_\odot$, the He-burning only lasts for a few My. The carbon core formed remains convective, and carbon burns to oxygen and magnesium. Further fusion synthesis occurs in several zones, leading to the production of elements up to ^{40}Ca, ^{44}Ti, ^{48}Cr, ^{52}Fe and ^{56}Ni, partly by He-capture partly by direct fusion of heavier nuclides. The heaviest elements may be formed in reactions like

$$^{28}\mathrm{Si} + {}^{28}\mathrm{Si} \rightarrow {}^{56}\mathrm{Ni}\left(\beta^+, 6.1\ \mathrm{d}\right) {}^{56}\mathrm{Co}\left(\beta^+, 77.3\ \mathrm{d}\right) {}^{56}\mathrm{Fe}\left(\mathrm{stable}\right) \tag{12.32}$$

From Figure 4.3 it is seen that formation of elements higher than those of A around 60 through fusion processes are exoergic (i.e. requires energy). When fusion processes end the star consists of an iron core surrounded by shells with silicon, oxygen, carbon, helium and hydrogen. Material is continually lost to space, propelled by strong solar winds. Massive stars of 20 $\mathbf{M}_\odot$ lose as much as 1/100 000 of their mass every year. The outer layer may have a temperature of only a few million degrees, while the center may be 10^9 K. In the lover density outer layer we might expect to find Li, Be, and B. However, these elements have been consumed in reactions with lighter nuclides.

The last steps of production of heavy elements (up to Fe/Ni) occurs rather rapidly in a few thousand years. When the nuclear fuel for fusion is exhausted the star collapses and results in a *supernova*. Figure 12.9 illustrates the composition of a 20 $\mathbf{M}_\odot$ star just before a supernova explosion.

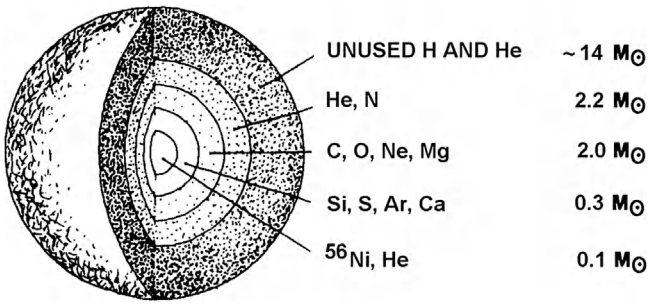


Figure 12.9 Shell structure of a 20 $M_\odot$ star just before supernova explosion.

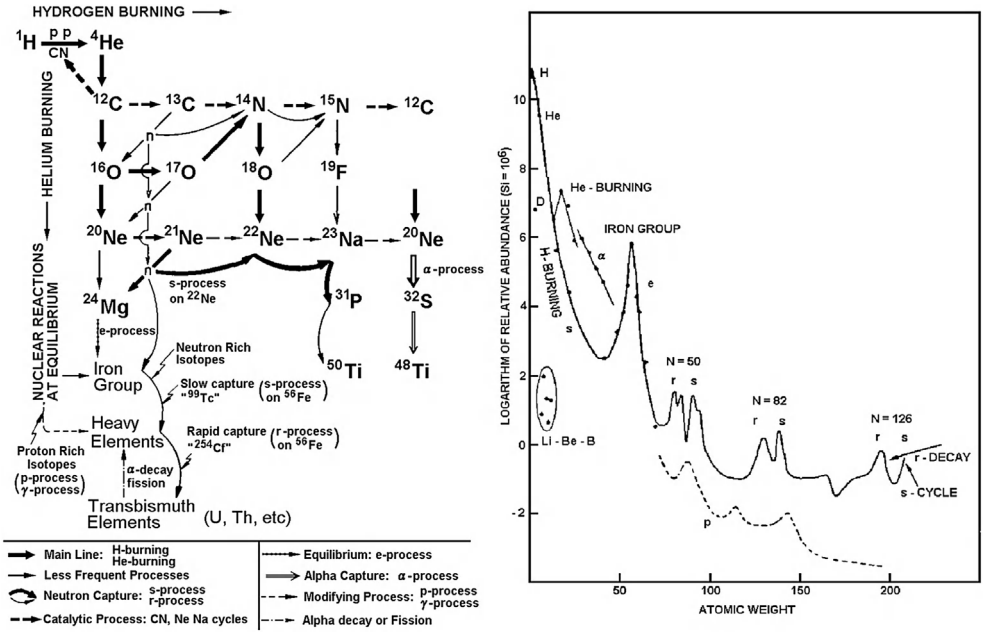


Figure 12.10 a. Schematic diagram of the nuclear processes for synthesis of the elements in stars. b. Calculated atomic abundances. *(From Burbidge, Burbidge, Fowler and Hoyle.)*

The sequence of element formation is summarized in Figure 12.10a (from the famous paper by Burbidge, Burbidge, Fowler and Hoyle). Figure 12.10b is from same paper and shows the calculated elemental composition, which is not too different from that in Figure 12.5a.

The short half-life of ^{8}Be plays a significant role. If ^{8}Be was much more stable, the helium-consuming chain would have proceeded much more rapidly. In fact after the helium ignition the energy production rate would have increased enormously and the red giant would have exploded as a supernova. However, the burning would not have gone much further than to ^{12}C. On the other hand, was ^{8}Be less stable, the fusion synthesis would never have been able to bridge the mass 8, and no elements higher than Be would have been formed.

The *Q*-value for the reaction $^{4}\text{He} + {}^{12}\text{C} \rightarrow {}^{16}\text{O}$ is 7.16 MeV. The excited levels of ^{16}O near this value are 7.12 and 8.87 MeV. The 8.87 MeV level is not useful for ^{16}O production since the Boltzmann distribution provides too small a fraction of particles with the needed kinetic energy. However, the resonance width of the 7.12 MeV level makes the reaction possible for the high energy tail of the Boltzmann distribution. Had it been easier to make ^{16}O, we would have had less carbon and more oxygen in the

universe. This probably would have been a hindrance in the development of life, since it is believed that life as we know it must start in a reducing (low oxygen containing) atmosphere.

Our galaxy has stars much larger and much smaller than our sun, and of widely different ages. Thus the processes we have described are occurring at present in the Milky Way and in other galaxies as well. Moreover, the stars emit considerable amounts of matter into space. As a result, interstellar gas, out of which new stars are formed, contains atoms heavier than helium, although hydrogen is the most abundant element.

12.5. NEUTRON CAPTURE PROCESSES: FROM IRON TO URANIUM

Figure 4.3 shows that the maximum nucleon binding energy occurs at $A \approx 60$, i.e. around iron, which we may consider to be the most thermodynamically stable element in the universe. At lower values of A, fusion of lighter elements releases energy while the exothermic reactions to form heavier elements ($A > 60$) involve neutron capture.

12.5.1. Slow neutron capture

Through hydrogen and helium burning, neutrons are formed. The most important reaction is believed to be

$$^{22}\mathrm{Ne} + {}^{4}\mathrm{He} \rightarrow {}^{25}\mathrm{Mg} + \mathrm{n} \tag{12.33}$$

As the heavier elements form in the star, the neutron production increases considerably since such reactions become more prevalent as heavier elements are involved in the reactions. In reactions like (12.31) and (12.32) energetic γ-rays are emitted, which decompose D into H and n (the binding energy is 2.22 MeV). The mode of production of the elements changes from that of helium capture to that of neutron capture, so that the elements from iron to bismuth can be formed by a slow process of neutron capture (n,γ-reactions), interrupted by β-decay whenever it is faster than the next capture step.

Such a process is known as the *slow*, or *s-process*. While the reaction probability for the capture of neutrons increases with the atomic number of the element, the relative amount of the elements in the star will decrease with increasing atomic number, because of the successive higher order of reaction (§17.3). The result is the observed flattening of the abundance curve for $A > 100$, see Figures 12.5a and 12.10b.

The formation of ^{104}Pd from ^{100}Ru can serve as an example of the steps in the s-process of element formation